Ray Tracer Simulation

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I. Introduction

By invoking Snell's law and taking geometric calculations in Cartesian coordinates, I constructed a computational ray tracer to investigate the imaging property of spherical surfaces. Ray optics was assumed, with no consideration of wave and Fourier optics. Using the ray tracer, plano-convex singlet lens was approximated. I also attempted to build a best-form lens algorithm that optimised surface curvatures.

II. Spherical Refraction

I began with a class representing a ray with position p and direction k variable attributes, and a class representing a spherical surface with its centre lying on the z-axis. To calculate the position of the intersection, if any, the equation provided by the project script [1] was used:

$$l\hat{\mathbf{k}} = \left[-(\mathbf{r}\cdot\hat{\mathbf{k}}) \pm \sqrt{(\mathbf{r}\cdot\hat{\mathbf{k}})^2 - (|r|^2 - R^2)}\right]\hat{\mathbf{k}} + \mathbf{P}, \quad (1)$$

where l would be the scalar factor of the unit directional vector that propagates the ray from its source P to the surface. There were three different cases of solutions with eq[1]: complex roots, which suggested that there was not an intersection; repeated real root, which suggested that the ray was tangent to the surface; and two real roots, where only one of them was possible as the surface was at most a hemisphere. To decide for the last case, a comparison was made to show which root's z value lies between z_0 and that of the endpoint of the surface. Because of the squared root operation in eq[1], a floating point error could be encountered when comparing two supposedly equal floats. Tolerance was given when comparing almost equal float to mitigate this error, using the numpy.isclose() function.

Knowing the position of the intersection and thus the direction of the normal $\hat{\mathbf{n}}$, Snell's law could be used to calculate the direction of the refracted ray $\hat{\mathbf{k_2}}$ from that of the incident ray $\hat{\mathbf{k_1}}$, assuming the surface was a boundary between regions of different refractive indices n. Snell's law function was constructed based on its vector formulation sourced form [2]:

$$\hat{\mathbf{k_2}} = \sqrt{1 - \left[1 - \mu^2(\hat{\mathbf{n}}\cdot\hat{\mathbf{k_1}})^2\right]}\hat{\mathbf{n}} + \mu\left[\hat{\mathbf{k_1}} - (\hat{\mathbf{n}}\cdot\hat{\mathbf{k_1}})\hat{\mathbf{n}}\right], \quad (2)$$

where $\mu = \frac{n_1}{n_2}$. Subscripts 1 and 2 denoted incident and refracted respectively. No refraction would occur if the following was satisfied:

$$\sin(\theta_1) > \frac{n_1}{n_2}, \quad (3)$$

where θ was the angle between the normal and ray. Total internal reflection would occur instead. Fig.1 shows a plot of a wave propagated from the origin, through a positive spherical surface. $\frac{n_1}{n_2}$ was calculated to be approximately 1.5, which was equal to $\frac{\sin\theta_2}{\sin\theta_1}$. This agreed with Snell's law to 15 significant figures.

III. Uniform Collimated Beam

A uniform collimated bundle of rays would be needed for later investigation with focal lengths and image size. Thus, I constructed a new class, containing an array of objects of the class Ray, using composition. The rays were arranged in a concentric circular pattern, with equally spaced rings of rays surrounding a central ray. The numbers of rays per ring were $6N$, where N was an integer representing the ring number. Rays within each ring were also equally spaced. Thus, the density would be approximately uniform throughout the beam's cross-sectional area.

The beam of radius 10 mm was propagated through a spherical surface (see Fig.2). Then it was projected onto a screen on the xy plane represented by another class. Intersection between the screen and each ray was computed to give an output image plot, as shown in Fig. 3. As expected, the beam converged to a focal point on the z-axis, before diverging again. Because of spherical abbreviation, the rays were shown not to be uniformly distributed around the focal point. Ring of rays of different radii converged to a single point at a slightly different value of z depending on the radius.

IV. Focal Length & Image Size

A. Point Separation

The image size, which was the distribution of points projected by the beam on the output screen, was investigated. The diameter of each ring was calculated, by implementing a method of the class Beam to calculate the separation between opposite pairs of points lying on the x-axis.

50 discrete samples between 195 mm and 203mm were taken by adding screen objects between the output screen and the refractive surface, at constant intervals around where the beam converged. For each ring, separations between opposite points were plotted against z (see Fig. 4a). Absolute linear function $y = |x|$ was fitted to find the value of z where separation was 0.

Focal length could be estimated by finding the mean, with the standard deviation approximated by halving the range. For a surface of $z_0 = 100$, $K = 0.03$ mm^{-1}, $n_1 = 1, n_2 = 1.5$ and $r_a = \frac{1}{0.03}$ mm, $f = (199.110.98)$ mm was calculated.

TABLE I
PLANO-CONVEX SINGLET LENS PERFORMANCE

Orientation	f /mm	r_f /mm
1	197.556971 ± 2×10^{-6}	0.0242574 ± 2×10^{-7}
2	198.16659564 ± 3×10^{-8}	0.097958860 ± 2×10^{-9}

B. RMS Spot Radius

Alternatively, one could estimate the focal length by calculating the root mean squared (RMS) spot radius of each point from the z-axis instead. For a uniform collimated beam centred around the z-axis, the image also had a circular symmetry centred around the z-axis. Therefore, the RMS spot radius for each discrete sample could be calculated from separations computed previously. A hyperbole was fitted over the plot of radius against z in Fig. 4b, as given by [3]:

$$\frac{(y-j)^2}{b^2} - \frac{(z-i)^2}{a^2} = 1, \tag{4}$$

where a, b, i, j were arbitrary constant. The minimum point, which equalled $j+b$, gave the second approximation of focal point. The RMS spot radius r at focal point (see Fig. 5) could also be calculated and given i. For the same parameters, it was computed that $f = (198.39119993318 \times 10^{-10})$ mm and $r_f = (0.04253953593 \times 10^{-10})$ mm. The precision of f estimated by this method was significantly greater than that using separation only.

V. DIFFRACTION LIMIT

The resolution of a real imaging system using lenses is limited by diffraction [4]. A point object would produce an Airy pattern as an image, with intensity spread out over an area, characterised by a point spread function. By Rayleigh criterion, the minimal separation at focal length between two distinguishable images from two point objects is given by the Airy disc radius $1.22\frac{\lambda f}{d}$, where $\frac{\lambda f}{d}$ is the diffraction scale which could be extended to apply to non-point object.

For the ray tracer, resolution would only be applied for non-point object. And it would instead be limited by dimension of the object at a fixed object distance. The RMS spot radius could be used in the case of a beam. Taking the beam radius as the aperture radius, how diffraction scale and RMS spot radius change with beam radius could be plotted and compared. Fig. 6 was plotted using the same parameters as those used in the previous investigation. λ taken as 588 nm. It was shown that the two relationships were different, demonstrating that the resolving power of the ray tracer behaved differently compared to a real system. There was however an intersection, where the resolution limiting factors equal.

VI. PLANO-CONVEX SINGLET LENS

To simulate a plano-convex singlet lens, a new class, consisting of two SphericalRefraction objects, was constructed. One surface was a plane, with a curvature of 0. The second surface has positive curvature if it is facing the source (noted as orientation 1); a negative curvature if the lens was orientated in the opposite way (noted as orientation 2). Two surfaces were separated by a thickness parameter.

As an example to investigate, the curvature of the curved surface was set to 0.02 mm^{-1}, with a thickness of 5 mm and a refractive index of 1.5168. A beam of diameter 10 mm was propagated through by both orientations, as shown in Fig. 7. Table 1 showed the estimated focal length f calculated by RMS spot radius, and the RMS spot radius at focal length r_f, for both orientations of the lens. Orientation 1 resulted in a smaller f and a smaller r_f. The image would be sharper at focus.

VII. BEST FORM LENS

To simulate a best forms lens, a specific class was constructed. Similar to the PlanoConvex class, the BestFormLens class was a composition of two SphericalRefraction object. But both curvatures were unconstrained. A function using the class was then constructed, taking a pair of curvature stored in a (1,2) array as an input, and giving the RMS spot radius as an output. The aperture radius r_a, refractive indices n, separations between the source, lens and output screen were fixed. Using $scipy.optimize.fmint_nc$, the output of the function was minimised. A further constraint was placed on the curvatures: $\frac{1}{r_a} < |K|$. This ensured that surfaces would always intersect with the fix aperture radius. The surfaces were still allowed to cross over with one another however. Therefore, the actual aperture radius after optimisation may still be smaller than what was set initially.

A best-form lens was optimised using the same set of parameters as the test done with the plano-convex lens. z was set to 198 mm, equalled to the f of the plano-convex to the nearest integer for both orientations. Fig. 8 showed a plot of a 10mm beam propagated through the lens. The optimised curvatures were 0.0174 mm^{-1} and -0.0027 mm^{-1} An RMS spot size r of 0.044 mm was obtained. As r was minimised, the distance between the lens and the output plane would be r_f. Compared to the plano-convex singlet lens, the performance was better than orientation 2 where the plane surface was facing the source but worst than orientation 1 where the curved surface was facing the source instead. This suggested that further improvement could be made to the minimisation algorithm.

REFERENCES

[1] R. Kingham and C. Paterson, "Imperial college physics y2 computing - short projects (v2.4.1) project a: An optical ray tracer," 2013.
[2] "Snell's law in vector form," Oct, 2018. [Online]. Available: https://physics.stackexchange.com/questions/435512/snells-law-in-vector-form
[3] J. Redden, *8.4 Hyperbolas*, ser. Intermediate Algebra. Saylor Academy, Aug, 2012.
[4] M. Damzen, *Angular Resolution of Optical Systems*, ser. Year 1 "Optics" Course Notes. Imperial College London, pp. 50–51.

APPENDIX
FIGURES

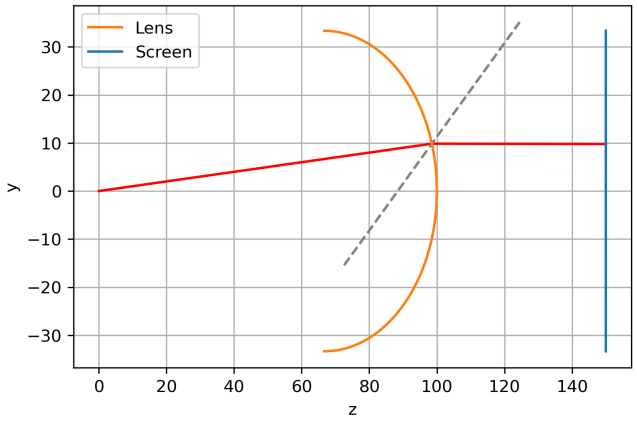


Fig. 1. A single ray (denoted by the red line) propagated through a spherical refractive surface at z_0 = 100 mm of K = -0.03 mm^{-1}, projected onto an output plane at z = 250 mm. The grey dotted line showed the surface normal at the intersection. The plot showed the y and z positions

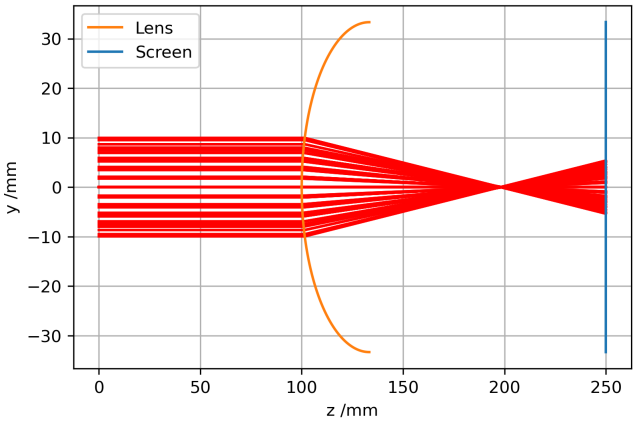
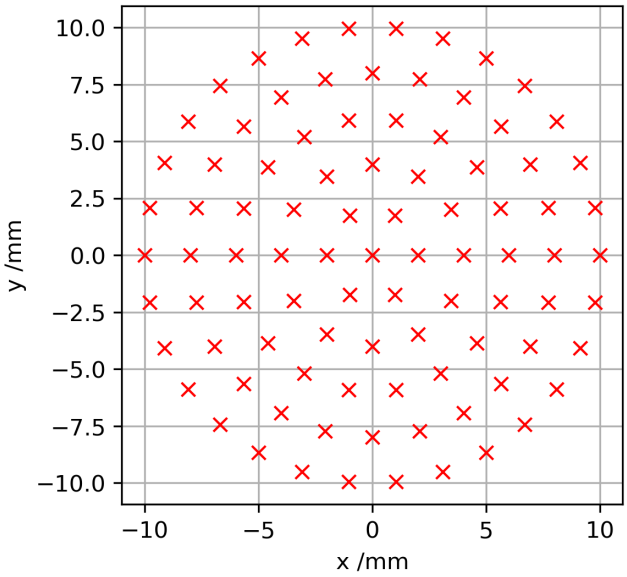
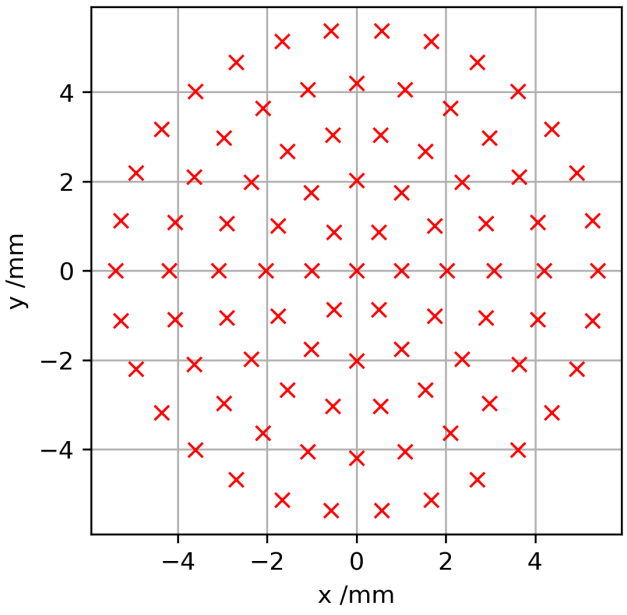


Fig. 2. A beam of diameter 10 mm propagated through a spherical refractive surface at z_0 = 100 mm of K = 0.03 mm^{-1}, projected onto an output plane at z = 250 mm. Beam converged at around z = 200 mm

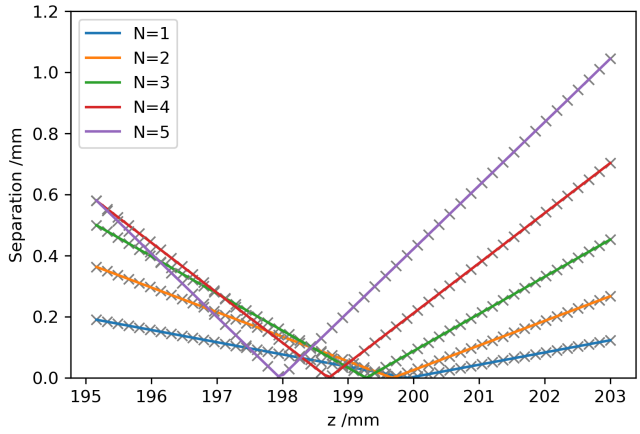


(a)

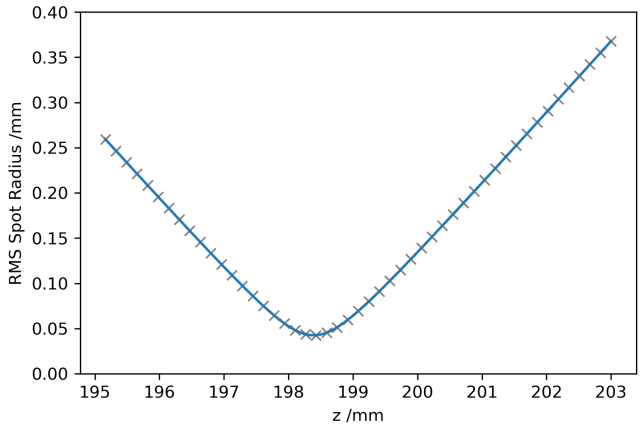


(b)

Fig. 3. (a) showed the initial position of a beam of diameter 10 mm. (b) showed the image of the beam after the propagation shown in Fig. 2



(a)



(b)

Fig. 4. (a) showed how ring diameters varies with z between 195 mm and 203 mm. N would be the index number of the ring, which 1 being the closest to the centre and 5 being the outermost. (b) showed how the RMS spot radius r varied with z within the same interval.

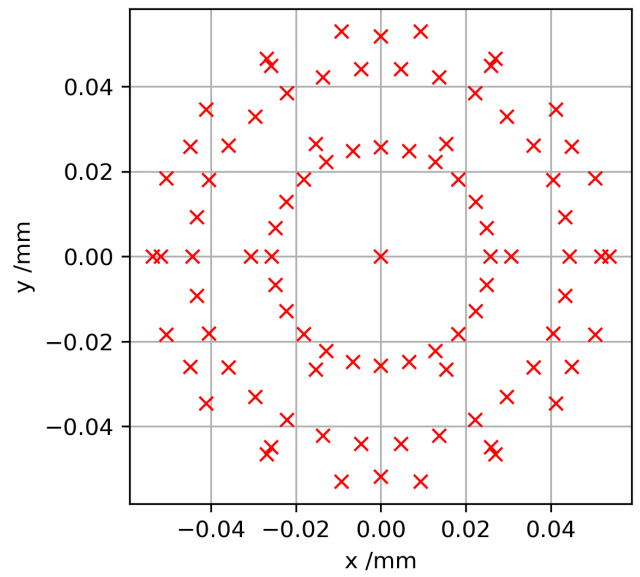


Fig. 5. The image projected at focal point f for the propagation shown in Fig. 2 was shown.

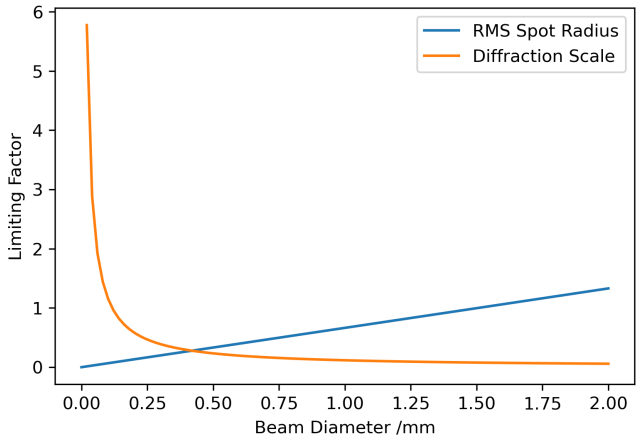
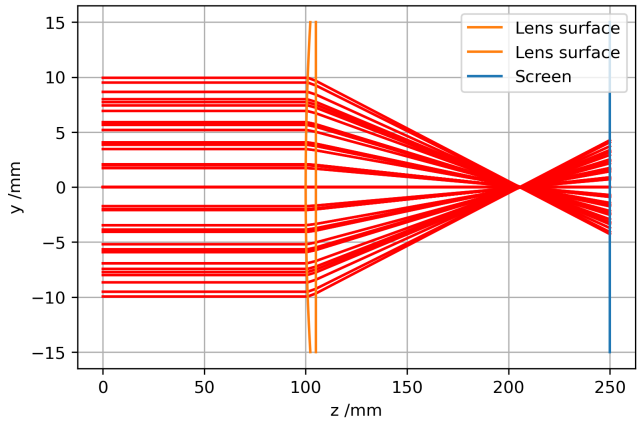
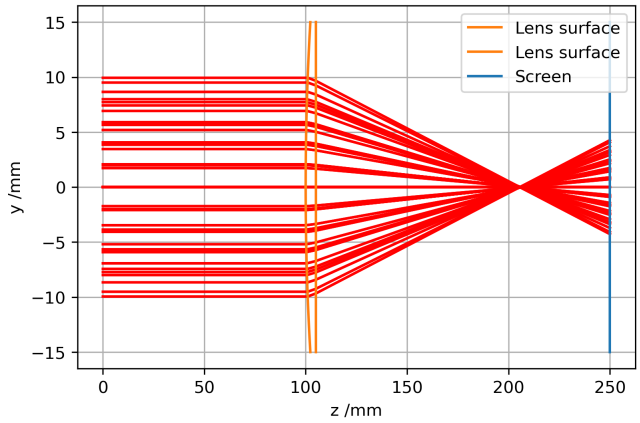


Fig. 6. The relationship between beam diameter and two different resolution limiting factors was shown. RMS spot radius was the factor for the ray tracer, while the diffraction scale was the factor for a real imaging system affected by diffraction.



(a)



(b)

Fig. 7. The propagation of a 10 mm beam through a plano-convex singlet lens was shown. (a) showed orientation 1, with the curved surface of K = 0.02 mm $^{-1}$. (b) showed the opposite setup, denoted as orientation 2.

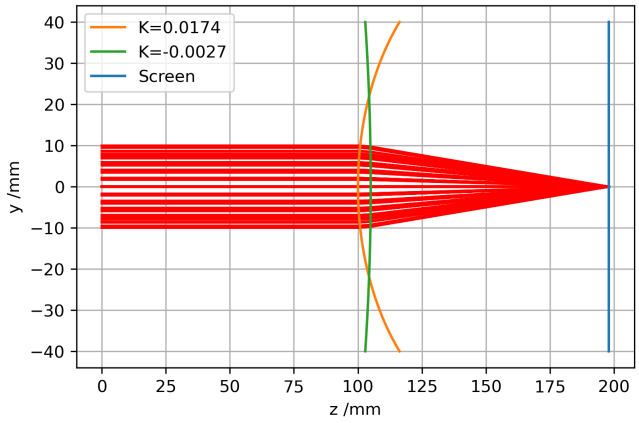


Fig. 8. The propagation of a 10 mm beam through a best-form lens optimised with the algorithm